

Does a Governing Equation Help? A Controlled Evaluation of a Convection-Diffusion-Reaction Physics-Informed Neural Operator for Forest Fire Susceptibility Mapping in India

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Abstract—Embedding a governing equation in a learning model is widely expected to improve environmental prediction, particularly when test conditions differ from training conditions. For forest fire susceptibility this expectation has not been examined under controlled conditions. We built a physics-informed Fourier neural operator whose latent field obeys a convection-diffusion-reaction (CDR) equation, with diffusion, advection and reaction coefficients driven by vegetation and moisture, terrain, and access covariates, and trained it on monthly MODIS fire detections over India from 2000 to 2022. The model was compared with three controls: the same network trained without the equation, classical models fitted on exactly the same cells, splits and covariates, and baselines that use no covariates at all. The comparison covered random, spatial block, held-out region and held-out year evaluation, three random seeds and paired significance tests. No physics configuration improved on the network without physics. On the random split the full CDR model reached a ROC-AUC of 0.939 against 0.945 without the equation (DeLong $p < 0.02$ for every seed), and on spatial and regional hold-out the differences could not be distinguished from zero. On identical 12 km cells and covariates a random forest scored 0.980, 0.974 and 0.959 on the random, spatial and regional tracks, compared with 0.939, 0.719 and 0.570 for the operator. For held-out years the operator (0.893) scored below a climatological fire frequency baseline (0.908) and a seasonal frequency baseline (0.930). Diagnostics of the trained field showed a nearly static solution whose equation residual was 74 to 616 times its own rate of change. The analysis rests on a rebuilt national pipeline with 541 545 forest fire points, 55 predictors on a 1/120 degree grid, a corrected fire label and seasonal trend statistics, in which a random forest reaches 0.975 over all pixels and 0.897 over forest pixels. The accuracy differences observed in this problem are explained by the model family and the predictor set, not by the physics.

Index Terms—Forest fire, susceptibility mapping, physics-informed machine learning, Fourier neural operator, convection-diffusion-reaction, spatial cross-validation, MODIS, India.

I. INTRODUCTION

FOREST fires are a recurring hazard across India, and roughly a third of the national forest area is classed as fire prone in official assessments cited by recent work [1]. Maps that rank locations by their long-term likelihood of burning

are used to plan patrols, fuel treatment and early warning. The most recent national map was produced by Biswas *et al.* [1], who trained a maximum entropy (MaxEnt) model on fifteen static predictors at 0.25 degree resolution and reported a test area under the receiver operating characteristic curve of 0.879.

Regional studies in India use a wide range of learners. Examples include analytic hierarchy weighting combined with tree ensembles in Goa [2], a six-model ensemble for the Western Ghats [3], MaxEnt in Tamil Nadu [4], boosted and bagged trees in Odisha [5], six learners in Mizoram [6], stacked ensembles in the northeast [7], explainable ensembles in the Western Himalaya [8] and fuzzy weighting in Jammu and Kashmir [9]. Studies elsewhere follow the same pattern, with convolutional networks in Yunnan [10], random forests and neural networks in Turkey [11], [12], gradient boosting in New South Wales and Greece [13], [14], evidential learning in Iran [15] and driver analysis in Portugal [16]. Nearly all of this work treats susceptibility as a static classification of locations and reports a single random split, although spatially blocked validation is recommended whenever the data are spatially autocorrelated [17].

Physics-informed neural networks add the residual of a differential equation to the training loss [18], and neural operators learn mappings between whole fields on a grid [19]. Fourier neural operators have been applied to climate downscaling [20], hydrological ensembles [21] and global weather forecasting [22]. The arguments usually made for physics-informed learning are better behaviour when data are scarce or shifted, physically consistent outputs and mechanistic interpretability [23], and process guidance has been shown to help lake temperature prediction when training data are thinned [24]. In the fire domain, physics-informed methods have so far addressed fire spread and data assimilation for individual events [25], [26], not susceptibility over a country.

The open question is therefore not whether a physics-informed susceptibility model can be built, but whether the equation contributes anything once other factors are held fixed. These factors include the capacity of the network, the choice of predictors, the grid and evaluation population, and the structure of the test split. Separating them requires controls that are rarely reported together: the same network without the equation, classical learners given identical inputs, and baselines that use no covariates. We address four research questions.

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The code, result files and per-cell predictions used in this study are available in the project repositories listed in the Data Availability section.

- **RQ1.** Do diffusion, advection and reaction terms improve held-out discrimination relative to the same network trained without them, under random, spatial, regional and temporal splits?
- **RQ2.** How does the operator compare with classical models trained on the same cells, partitions and covariates?
- **RQ3.** For held-out years, does the operator outperform baselines that ignore covariates?
- **RQ4.** Does the trained operator behave as a solution of its own equation?

The contributions of this paper are as follows.

- 1) A controlled test of a physics-informed operator for fire susceptibility: four physics configurations, four evaluation tracks and three seeds, with paired tests, same-cell classical comparisons and persistence baselines. The answer to RQ1 to RQ3 is negative, and a paired decomposition locates where the accuracy differences actually arise.
- 2) A physical consistency diagnostic based on term magnitudes and the equation residual of the trained field, which shows that a physics-regularised model can score well while its field neither evolves in time nor satisfies the equation (RQ4).
- 3) A reproducible national pipeline for India in which every statistic was recalculated from raw data. It corrects a half-pixel misregistration of fire points, replaces degenerate anomaly predictors with climatological levels, replaces Mann-Kendall tests on seasonal series with the Seasonal Kendall test, restricts spatial autocorrelation to cells inside the country and uses land cover from before the label period. Forest pixels are evaluated as the primary population.
- 4) A reimplement of the reference MaxEnt protocol on the same data, which places the reference result in context and states which pixel-level scores can and cannot be compared with it.

The paper is organised as follows. Section II describes the study area, data and predictors. Section III presents the classical models, the operator and the evaluation design. Section IV reports the results, Section V interprets them and Section VI concludes.

II. STUDY AREA AND DATA

A. Domain, Period and Grid

The study area is India, delimited by a dissolved state boundary polygon reprojected to geographic coordinates (EPSG:4326). The study period runs from 1 November 2000 to 15 December 2022 (266 months), bounded by the availability of annual ESA CCI and C3S land cover maps. All raster products are aligned to a common grid of 3641×3504 cells with a spacing of $1/120$ degree (about 0.93 km). The India mask contains 4 184 671 cells, of which 4 160 768 have valid vegetation data and form the analysis population. Cells with a 2001 forest fraction above zero (1 197 538 cells) form the forest population.

B. Fire Observations and Label

Active fire detections were taken from the MODIS Collection 6.1 archive [27]. Clipping to a bounding box, clipping to the India polygon, removing duplicates by location and date, and keeping only detections that fall on a forest class of the same year's land cover map (thirteen forest codes) reduced 2 804 373 records to 541 545 forest fire points. Detections with confidence below 30 (4.29%) and non-vegetation type codes (0.21%) were retained, because excluding either group changed ROC-AUC by less than 0.001. Annual counts agree with those implied by the published shares of [1] to within 0.5% to 2.4% ($r = 0.99996$, 2001 to 2020). Forest cover inside the Indian boundary ranges from 18.3% to 19.3% over the period.

A cell is labelled positive if it contains at least one forest fire point during the study period. Each point is assigned to the cell that contains it,

$$c = \left\lfloor \frac{\lambda - x_0}{\Delta x} \right\rfloor, \quad r = \left\lfloor \frac{\varphi - y_0}{\Delta y} \right\rfloor, \quad (1)$$

where (x_0, y_0) is the outer corner of the grid, (λ, φ) are the point coordinates and $\Delta y < 0$. The label has 268 411 positive cells, a prevalence of 6.45% over all cells and 22.4% over forest cells. Rounding instead of flooring in (1) measures from the cell edge rather than the cell centre and moves 74.9% of points into a neighbouring cell. Correcting it raises random forest ROC-AUC by 0.006 over all pixels and 0.011 over forest pixels.

Fig. 1 shows the spatial distribution of the 541 545 detections before rasterisation. Fires concentrate in three areas: the central Indian highlands and Eastern Ghats, the northeastern hill states, and the Himalayan foothills, with comparatively few detections in the arid northwest and the humid southwestern coast.

C. Predictors

The 55 predictors are summarised in Table I. They cover all fifteen variable groups of [1]. Climate and land surface temperature enter as 2001 to 2020 climatological levels together with a Seasonal Kendall trend statistic. Land cover fractions are taken from the 2001 map so that they precede almost the whole label period. Three design points deserve comment. First, a time mean of anomalies computed against a 2001 to 2020 baseline equals the residue of the 26 months outside the baseline, so it carries almost no spatial information: the spatial standard deviation of each climatological level is 40 to 431 times that of its anomaly mean. Second, five variable groups come from products that differ from those of the reference study (MOD11A2 for land surface temperature, MOD13A3 for vegetation, and FLDAS for precipitation, soil moisture and net longwave radiation). Third, distances were validated against exact geodesic distances at 3000 random locations, with root mean square errors of 0.31, 0.65 and 0.32 km for roads, railways and waterways.

Fig. 2 shows the dominant class of the 2001 land cover map used for the land cover group, grouped into eight categories for legibility. Cropland covers most of the Indo-Gangetic plain,

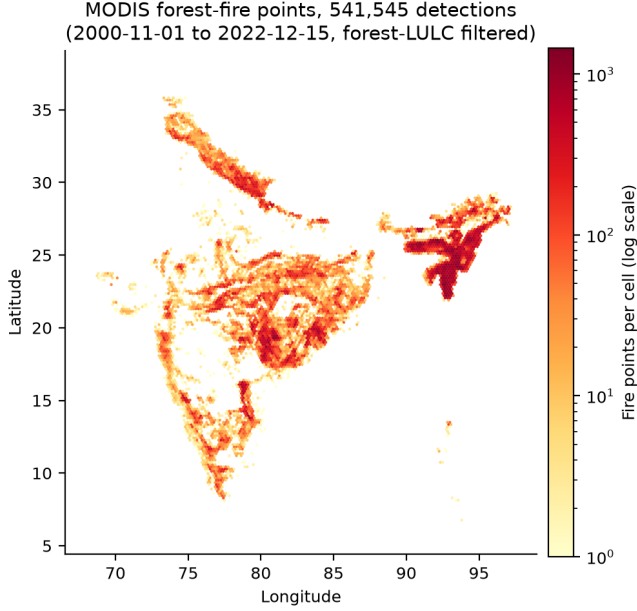


Fig. 1. Spatial density of the 541 545 MODIS forest-fire detections used in this study, before assignment to the analysis grid.

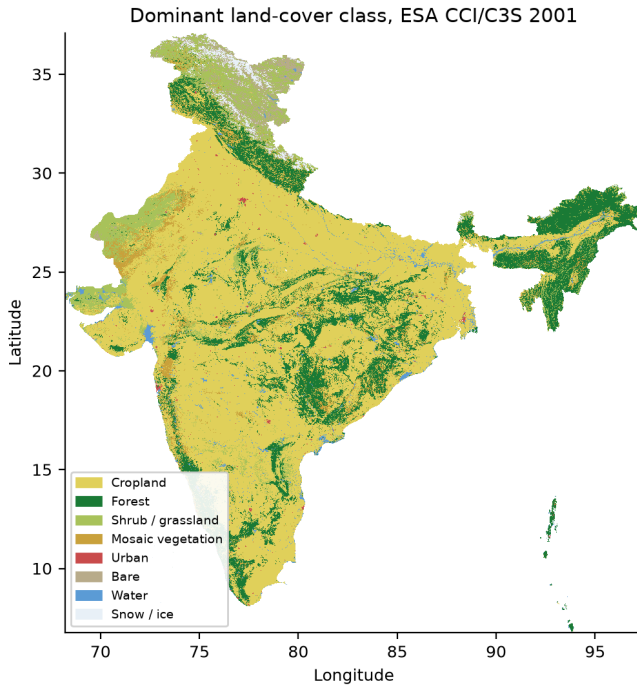


Fig. 2. Dominant land-cover class of India from the 2001 ESA CCI/C3S map, grouped into eight categories from the 22 classes underlying the 21 land cover fractions in Table I.

forest is concentrated in the central highlands, the Western Ghats and the northeast, and the Himalayan belt is a mix of forest, shrubland and grassland by elevation.

TABLE I
PREDICTOR SET USED BY THE CLASSICAL MODELS (55 FEATURES)

Group (count)	Source	Features
Vegetation (6)	MOD13A3.061, 1 km, monthly	QA-masked mean NDVI; climatological June NDVI; Seasonal Kendall τ ; seasonal Sen slope; CVSI at lag $k^* = 8$; LISA cluster class
Surface temp. (6)	MOD11A2.061, 1 km, 8-day	Climatological level and Seasonal Kendall τ of day LST, night LST and diurnal range
Climate (14)	FLDAS Noah, 0.1°, monthly	Level and Seasonal Kendall τ of air temperature, specific and relative humidity, wind, precipitation, net long-wave radiation and soil moisture
Terrain (4)	SRTMGL3, 90 m	Elevation; Horn slope [28]; sine and cosine of aspect
Access (3)	OpenStreetMap 2022	Distance to roads, railways and waterways
Land cover (22)	ESA CCI/C3S 2001	21 class fractions and forest fraction

III. METHODS

Fig. 3 gives an overview of the workflow, from the raw products to the two model branches and the evaluation stage.

A. Trend and Spatial Statistics

Per-cell trends were tested with the Seasonal Kendall statistic [29], with tie correction and at least 30 comparable pairs, combined with the seasonal Sen slope [30] and Benjamini-Hochberg control of the false discovery rate at $q < 0.05$ [31]. The Mann-Kendall test was not used because monthly series are strongly seasonal, and a moving-average series of NDVI used in earlier work had a lag-one autocorrelation of 0.975. Global Moran's I and local indicators of spatial association [32] were computed on 8×8 block means restricted to cells inside India, with 999 permutations.

B. Classical Susceptibility Models

A random forest [33] with 200 trees, a maximum depth of 25, a minimum leaf size of 3 and balanced class weights, and a MaxEnt model [34] with linear, quadratic, hinge and product features and a regularisation multiplier of 4.0 were trained on a stratified 65/15/20 split into training, validation and test cells. Hyperparameters were selected on the validation set, imputation used training rows only, decision thresholds were chosen by maximum F1 on validation, and the test set was scored once. MaxEnt was fitted on 150 000 training rows, and a sensitivity analysis from 50 000 to 500 000 rows is reported. Both learners were also trained on the fifteen predictors of [1], expressed as levels, to separate the effect of the predictor set from the effect of the model family.

C. Convection-Diffusion-Reaction Operator

1) *Governing equation:* Let $u(\mathbf{x}, t)$ be a latent logit field on the domain Ω with susceptibility $s = \sigma(u)$, where σ is the logistic function. The operator is regularised towards

$$\frac{\partial u}{\partial t} = D(\mathbf{x}, t) \nabla^2 u - \mathbf{v}(\mathbf{x}) \cdot \nabla u + \rho(\mathbf{x}, t) \sigma(u) (1 - \sigma(u)) \quad (2)$$

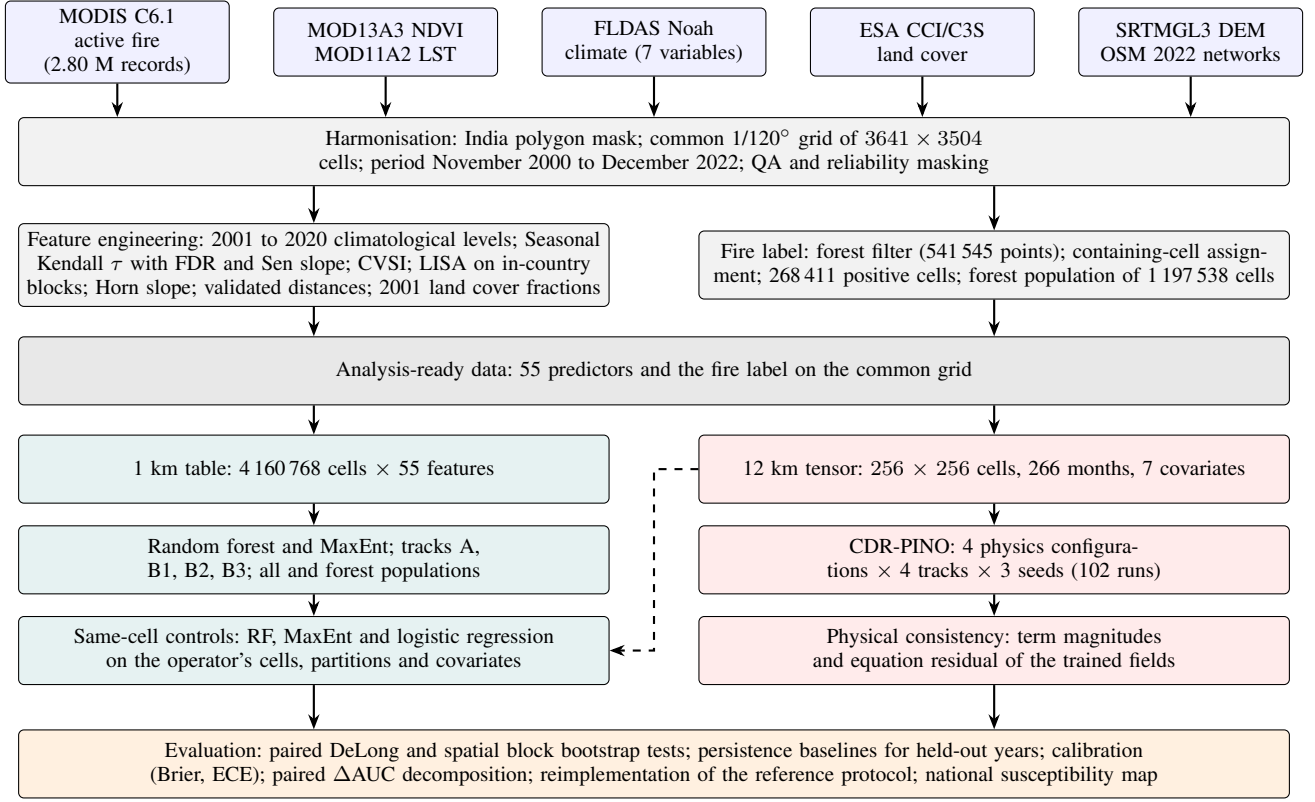


Fig. 3. Workflow of the study. Blue boxes are raw data products, grey boxes are shared preprocessing, green boxes form the classical branch at 1 km, red boxes form the operator branch at 12 km and the orange box is the common evaluation stage. The dashed arrow indicates that the same-cell controls are fitted on the operator's own tensor, cells and partitions.

on $\Omega \times (0, T]$, with a zero-flux boundary condition $\partial u / \partial n = 0$ on $\partial \Omega$ and the initial state $u(\mathbf{x}, 0) = 0$. The three coefficients are produced by small networks:

$$D = \text{softplus}(f_D(\text{NDVI}, F) - \text{softplus}(w) \text{NDVI}'), \quad (3)$$

$$\mathbf{v} = \text{softplus}(c) \nabla E, \quad (4)$$

$$\rho = \text{softplus}(f_\rho(Q, \text{NDVI}, S, R)), \quad (5)$$

where F is the 2001 forest fraction, NDVI' the monthly NDVI anomaly, E elevation, Q a monthly dryness index, S slope and R distance to roads. The functions f_D and f_ρ are multilayer perceptrons with two hidden layers of 12 units and hyperbolic tangent activations, and w and c are scalar parameters. Diffusion is thus tied to vegetation and moisture, advection to upslope transport along the elevation gradient, and reaction to ignition pressure. Because the reaction acts on the logit, in terms of s it reads $ds/dt = \rho s^2(1-s)^2$, so it is not a Fisher-KPP term in s . For bounded coefficients with $D \geq D_{\min} > 0$, uniform parabolicity, a Gårding inequality for the advective term and a reaction that is globally bounded by $\rho_{\max}/4$ and globally Lipschitz with constant $\rho_{\max}/(6\sqrt{3})$ give global existence and uniqueness for (2). These properties concern the equation. Section IV-F tests whether the trained network satisfies it.

2) *Architecture*: The architecture is shown in Fig. 4. The operator $\mathcal{G}_\theta$ maps the current state and the covariates of month t to the next state, $u_{t+1} = \mathcal{G}_\theta(u_t, \mathbf{a}_t)$. The input has eight channels on a 256×256 grid (about 12 km, 22 542 valid

cells): the state u_t and seven covariates, namely mean NDVI, monthly NDVI anomaly, forest fraction, monthly dryness, slope, distance to roads and elevation. A pointwise convolution lifts the input to 32 channels. Four Fourier layers follow, each computing

$$\mathbf{z}_{\ell+1} = \text{GELU}(\mathcal{F}^{-1}(R_\ell \cdot \mathcal{F}(\mathbf{z}_\ell)) + W_\ell \mathbf{z}_\ell), \quad (6)$$

where $\mathcal{F}$ is the two-dimensional real Fourier transform, R_ℓ holds complex weights on the lowest 16×16 modes and W_ℓ is a pointwise convolution. Two pointwise convolutions with a GELU in between project back to one channel. The model has 1 054 613 trainable parameters, including the coefficient networks.

3) *Losses and training*: The residual of (2) is evaluated with a one-month forward difference and with spatial derivatives of the midpoint state $\bar{u} = (u_t + u_{t+1})/2$, computed by spherical spectral operators on a symmetric extension of the grid:

$$r = (u_{t+1} - u_t) - D \nabla^2 \bar{u} + \mathbf{v} \cdot \nabla \bar{u} - \rho \sigma(\bar{u})(1 - \sigma(\bar{u})). \quad (7)$$

The data loss is a class-weighted binary cross entropy over all training cells, mixed in equal parts with a cross entropy on a log-sum-exp pooled terminal state for the final window. The boundary loss penalises $|\nabla u|^2$ on the boundary ring as a proxy for zero normal flux. Loss weights are updated every five windows by $w_k \leftarrow 0.9 w_k + 0.1 \|\nabla \mathcal{L}\| / \|\nabla \mathcal{L}_k\|$, following the gradient balancing idea of [35]. Training uses truncated

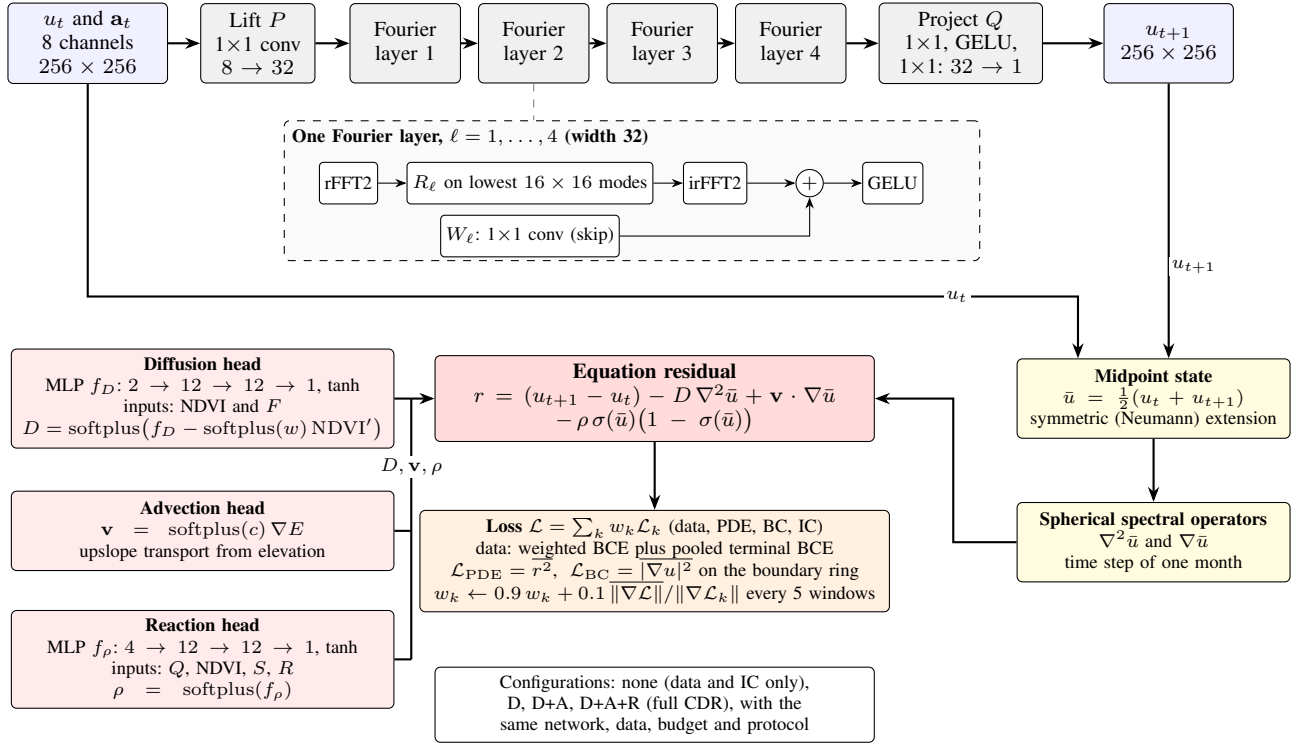


Fig. 4. Architecture of the CDR physics-informed neural operator. Top: the Fourier neural operator maps the state u_t and seven covariates of month t to u_{t+1} , with the structure of one Fourier layer shown in the inset. Bottom: three coefficient heads produce D , $\mathbf{v}$ and ρ , spherical spectral derivatives are evaluated on the midpoint state, and the equation residual enters the loss together with the data, boundary and initial terms under adaptive weights. The initial condition $u_0 = 0$ is imposed exactly, so $\mathcal{L}_{IC}$ is identically zero. The four physics configurations differ only in which terms enter the residual.

Algorithm 1: One training epoch of the operator

Input: monthly covariates $\mathbf{a}_{1:T}$, monthly fire indicators $\mathbf{y}_{1:T}$, training mask M , configuration $\kappa \in \{\text{none}, \text{D}, \text{D+A}, \text{D+A+R}\}$

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1  $u_0 \leftarrow 0$ ;
2 for each window  $[t_0, t_0 + 23]$  of 24 months do
3   detach  $u_{t_0}$  from the previous graph;
4   for  $t = t_0$  to  $t_0 + 23$  do
5      $u_{t+1} \leftarrow \mathcal{G}_\theta(u_t, \mathbf{a}_t)$ ;
6     accumulate  $\mathcal{L}_{\text{data}}$  on cells in  $M$ ;
7     if  $\kappa \neq \text{none}$  then
8       compute  $D, \mathbf{v}, \rho$  from the heads; accumulate  $\overline{r^2}$ 
        using the terms enabled by  $\kappa$ ; accumulate  $\mathcal{L}_{\text{BC}}$ ;
9   add pooled terminal cross entropy on the last state;
10   $\mathcal{L} \leftarrow \sum_k w_k \mathcal{L}_k$ ; update  $\theta$  with AdamW;
11  every 5 windows update  $w_k$  by gradient norm balancing;
12 every 5 epochs compute validation ROC-AUC; reduce the
    learning rate on a plateau of validation loss; stop after 4
    checks without improvement and restore the best
    checkpoint;
```

backpropagation over windows of 24 months. Algorithm 1 summarises one training epoch.

D. Controlled Evaluation Design

1) *Physics configurations*: Four configurations share the network, the data, the budget and the protocol: no physics (data and initial condition losses only), diffusion only, diffusion and advection, and the full equation.

TABLE II
EVALUATION TRACKS USED FOR ALL MODELS

Track	Held-out unit	Construction
A	random cells	65/15/20 split of cells
B1	spatial blocks	$2^\circ \times 2^\circ$ blocks in 3 folds; validation drawn as whole blocks from the training region
B2	regions	6 k -means regions, one held out at a time
B3	years	test years 2000, 2008, 2009, 2015; validation years 2001, 2012, 2013, 2020; scored per cell and month

2) *Evaluation tracks*: The tracks are listed in Table II. For the held-out year track, the terminal label used in training is rebuilt from the training months only, since pooling it over all years would expose the test period.

3) *Protocol*: Every operator run used AdamW with a learning rate of 10^{-3} and a weight decay of zero (selected on validation), a learning rate reduced by half after two validation checks without improvement in validation loss, up to 80 epochs, and early stopping on validation ROC-AUC checked every five epochs with a patience of four checks. Seeds 42, 43 and 44 change only the initialisation and training noise; fold, region and year assignments are fixed. In total 102 runs were trained.

4) *Same-cell controls and baselines*: Random forest, Max-Ent and logistic regression were fitted on the operator's own 12

km cells, partitions and seven covariates. A second set used 12 km aggregates of the 55 predictors, which separates the effect of the predictors from that of the learner. For held-out years, two baselines ignore covariates: the fire frequency of each cell over the training years, and its frequency in the same calendar month. A random forest on the monthly covariates, with and without the month as an input, is also reported.

5) *Statistics*: Paired differences on identical test units were tested with the DeLong method [36] on tracks A and B3. On tracks B1 and B2, where test cells are clustered in space, a spatial block bootstrap with 123 blocks and 300 resamples was used. Calibration is summarised by the Brier score and the expected calibration error (ECE). Because non-forest cells can never be positive under a forest fire label, forest fraction alone reaches an ROC-AUC of 0.91 over all cells. Every classical result is therefore reported for all cells and for forest cells, with the forest population treated as primary, and each ROC-AUC is stated with its grid and prevalence.

E. Physical Consistency Diagnostic

For trained checkpoints (track A, seed 42, all configurations) we computed, over all valid cells and months, the median of $|\partial u / \partial t|$, the mean magnitude of each term on the right side of (2) and the root mean square of the residual (7). A field that solves its equation should have a residual that is small relative to its own rate of change, and its rate of change should be clearly non-zero, because fire occurrence in India is strongly seasonal.

F. Reimplementation of the Reference Protocol

The protocol of [1] was reimplemented on the present data with a 0.25° grid (4630 cells), the fifteen predictors as levels, presences from 2020 forest fires (1292 cells), MaxEnt with linear, quadratic, hinge and product features at a multiplier of one, and ten random 75/25 presence splits. The counts reported in [1] (1830 training and 609 test presences) imply a 75/25 split, although the text states 70/30. This is a reimplementation and not the reference result.

IV. RESULTS

A. National Patterns

Under Seasonal Kendall with false discovery rate control, NDVI increased significantly at 3 552 278 cells and decreased at 72 305 (median Sen slope $+0.0026$ per year). Daytime land surface temperature decreased at 2 435 163 cells (median -0.050°C per year), night-time temperature increased at 2 080 747 cells ($+0.024^\circ\text{C}$ per year), and the diurnal range narrowed at 3 273 301 cells (-0.077°C per year). Specific humidity increased at 26 373 of 29 056 climate cells, and air temperature changed significantly at 9 988 cells, 7 620 of them decreasing. The Mann-Kendall test applied to seasonal series had described the air temperature trend as noise, which the Seasonal Kendall result does not support. Global Moran's I of NDVI over cells inside India is 0.9456 ($z = 494$, $p = 0.001$). Fire points lie on steeper terrain than the country as a whole (mean slope 12.35° against 5.72°) and are 38.9% closer to roads and 64.4% closer to waterways than the national mean.

TABLE III
CLASSICAL MODELS ON THE 1 KM TEST SET (ALL CELLS / FOREST CELLS)

Model and track	ROC-AUC	Average precision
RF, 55 features, A	0.975 / 0.897	0.720 / 0.721
MaxEnt, 55 features, A	0.967 / 0.866	0.663 / 0.664
RF, reference 15 predictors, A	0.970 / 0.885	0.692 / 0.698
RF, 55 features, B1	0.959 / 0.838	0.582 / 0.583
MaxEnt, 55 features, B1	0.956 / 0.828	0.561 / 0.563
RF, 55 features, B2	0.935 / 0.782	0.412 / 0.413
RF, 55 features, B3	0.965 / 0.872	0.282 / 0.282
Persistence baseline, B3	0.807 / 0.744	0.131 / 0.146

Prevalence 6.45% (all) and 22.4% (forest). Standard deviations over folds: B1 0.014 / 0.039 (RF), 0.012 / 0.029 (MaxEnt); B2 0.036 / 0.034 (RF).

TABLE IV
OPERATOR ABLATION, MEAN $\pm$ SD OVER SEEDS AND FOLDS (12 KM CELLS)

Track	None	D	D+A	D+A+R
A	0.945 $\pm$ 0.002	0.924 $\pm$ 0.005	0.939 $\pm$ 0.001	0.939 $\pm$ 0.002
B1	0.724 $\pm$ 0.010	0.642 $\pm$ 0.013	0.730 $\pm$ 0.004	0.718 $\pm$ 0.007
B2	0.614 $\pm$ 0.021	n/a	n/a	0.570 $\pm$ 0.007
B3	0.904 $\pm$ 0.001	0.886 $\pm$ 0.006	0.894 $\pm$ 0.001	0.893 $\pm$ 0.003

D: diffusion, A: advection, R: reaction. Prevalence 42% (A, cells) and 2.46% (B3, cell-months). Full minus none, paired: A -0.005 , -0.007 , -0.006 (DeLong $p < 0.02$ each seed); B1 and B2 intervals include zero; B3 -0.010 , -0.009 , -0.014 .

B. Classical Models at 1 km

Table III lists the classical results. Moving from all cells to forest cells lowers ROC-AUC by 0.08 to 0.15 and sharpens the ranking of learners. The random forest leads MaxEnt by 0.008 over all cells and by 0.031 over forest cells (paired DeLong). Spatial and regional hold-out lower forest ROC-AUC to 0.838 and 0.782. MaxEnt is better calibrated than the random forest (ECE 0.015 and 0.054 against 0.064 and 0.219 for all and forest cells), so the random forest output is interpreted as a relative score rather than a probability.

C. RQ1: Physics Terms Against the Same Network Without Them

Table IV and Fig. 5 report the ablation. No configuration outperforms the network without physics on any track. Where the difference can be resolved (tracks A and B3), the physics terms lower the score, by 0.005 to 0.007 on track A (DeLong $p < 0.02$ in each seed) and by 0.009 to 0.014 on track B3. On tracks B1 and B2 the block bootstrap intervals include zero. Diffusion alone is the weakest configuration wherever it was trained, and adding advection recovers most of the loss. The same ordering holds on forest cells (track A: 0.910 without physics against 0.897 with the full equation; track B3: 0.848 against 0.833). The physics terms increase training time by a factor of 2.3 to 2.4. Rebuilding the earlier held-out-year label, which included test years, changes the score by only 0.0006.

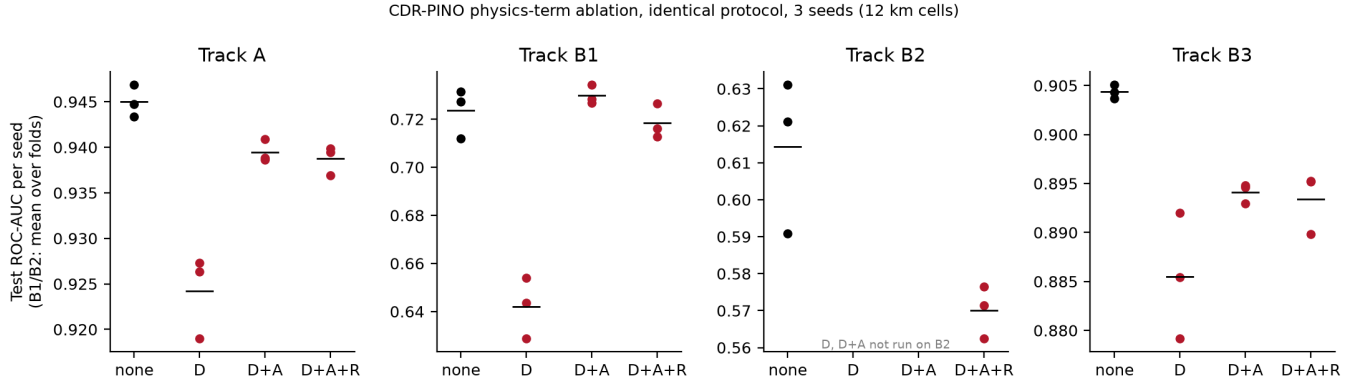


Fig. 5. Physics term ablation of the operator under an identical protocol. Dots are seeds (for B1 and B2, the mean over folds), horizontal lines are means. D, A and R denote diffusion, advection and reaction. Diffusion and diffusion with advection were not trained on track B2.

TABLE V
SAME 12 KM CELLS, PARTITIONS AND COVARIATES (ALL / FOREST CELLS)

Track	CDR full	CDR none	LogReg	MaxEnt	RF
A	0.939/0.897	0.945/0.910	0.924/0.874	0.958/0.927	0.980/0.950
B1	0.719/0.701	0.724/0.702	0.919/0.869	0.950/0.917	0.974/0.936
B2	0.570/0.569	0.614/0.635	0.872/0.835	0.845/0.824	0.959/0.916

B1 and B2 values are means over folds or regions.

D. RQ2: Classical Models on Identical Cells and Covariates

Table V and Fig. 6 compare the operator with classical learners given the same cells, partitions and covariates. The random forest leads by 0.041 on track A, and the gap grows to 0.26 on B1 and 0.39 on B2. Even logistic regression on the same seven covariates exceeds the operator by 0.20 on B1 and 0.30 on B2. Replacing the seven covariates by 12 km aggregates of all 55 predictors changes the classical scores by at most 0.01 (random forest 0.981, 0.976 and 0.953), so the covariates are not what limits spatial transfer.

E. RQ3: Held-Out Years Against Covariate-Free Baselines

On the four held-out years, scored per cell and month at a prevalence of 2.46%, the fire frequency of each cell over the training years reaches 0.908 (0.857 on forest cells), and the frequency in the same calendar month reaches 0.930 (0.920). The operator reaches 0.893 with the full equation and 0.904 without it. A random forest on the monthly covariates reaches 0.902, and 0.972 when the month is added (Fig. 7). The operator's skill on unseen years is therefore below what the recurrence of fire at a location already provides. Most of the temporal signal is location combined with season, which a model given the calendar month captures directly.

F. RQ4: Does the Trained Operator Solve Its Equation?

Table VI and Fig. 8 summarise the term magnitudes. In every configuration the field is almost static, with a median $|\partial u / \partial t|$ of at most 0.002 per month despite the strong seasonality of fire. The root mean square residual is 74 to 616 times

TABLE VI
TERM MAGNITUDES OF TRAINED TRACK A CHECKPOINTS (SEED 42)

Config.	Median $ \partial_t u $	Share of right side			Residual ratio
		Adv.	React.	Diff.	
None	0.0016	0.98	0.02	0.003	616
D	0.0003	0.83	0.16	0.002	224
D+A	0.0002	0.83	0.17	0.0004	74
D+A+R	0.0003	0.91	0.09	0.0004	78

Shares are fractions of the summed mean magnitudes of the three right-side terms, evaluated for every configuration with the full equation. Residual ratio: root mean square residual divided by the mean $|\partial_t u|$.

the mean rate of change, and diffusion contributes at most 0.3% of the right side. The optimiser meets the data loss with a nearly static map and leaves the residual unresolved. Advection carries 83% to 98% of the right side, which indicates that the advective head mainly routes elevation information into the model. Within the seven-covariate model, removing elevation had the largest effect in an earlier single-seed jackknife test, which was not repeated here. With the full predictor set, however, the terrain group is redundant (Table VII). The prominence of elevation is therefore a property of the restricted covariate set rather than of fire occurrence.

G. Sources of the Accuracy Differences

Table VII and Fig. 9 decompose the differences with paired comparisons. The largest effects come from the model family and from the predictor set. Restricting the random forest to the operator's five static covariates costs 0.017 over all cells and 0.071 over forest cells. Trend and land cover groups add small but positive amounts, the terrain group is redundant, and neither the physics terms nor the operator structure contribute positively.

H. Relation to the Reference Study

The reimplementing of the reference protocol reaches a test ROC-AUC of 0.893 ± 0.009 (range 0.877 to 0.912; training 0.902 ± 0.003), close to the 0.879 test and 0.894 training values reported in [1]. The leading permutation importances are reproduced: NDVI 19.0% (22.3% in the reference),

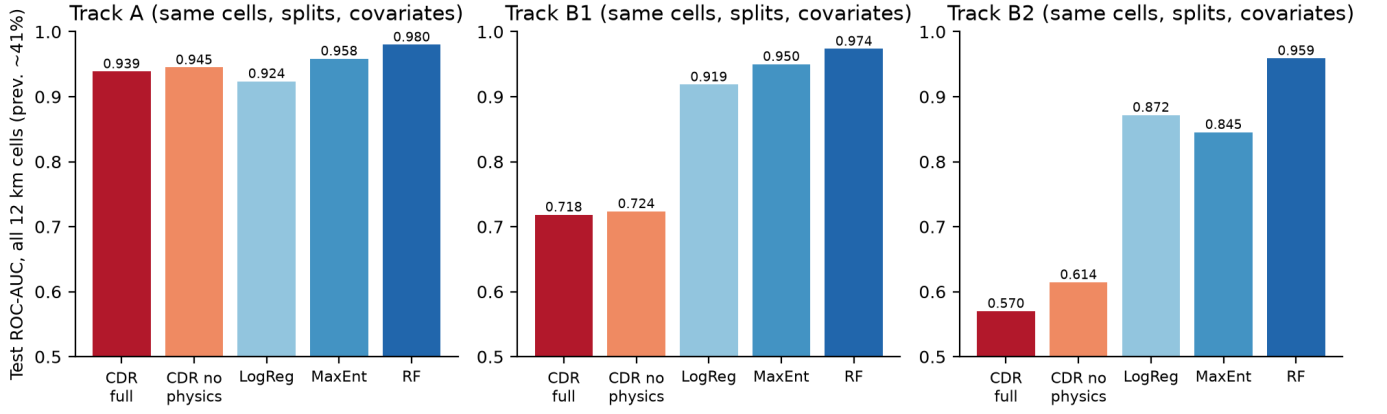


Fig. 6. Test ROC-AUC of the operator and of classical learners fitted on the same 12 km cells, partitions and seven covariates, for random cells (A), spatial blocks (B1) and held-out regions (B2).

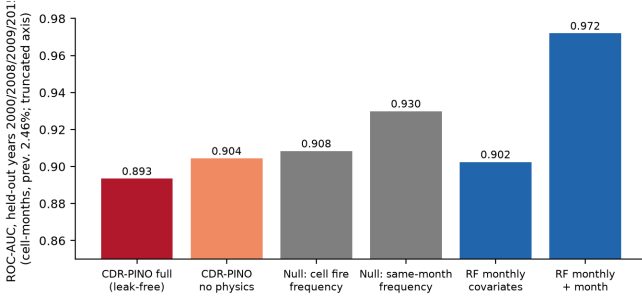


Fig. 7. Held-out years (2000, 2008, 2009, 2015) scored per cell and month. The operator is compared with two covariate-free baselines and with random forests on monthly covariates.

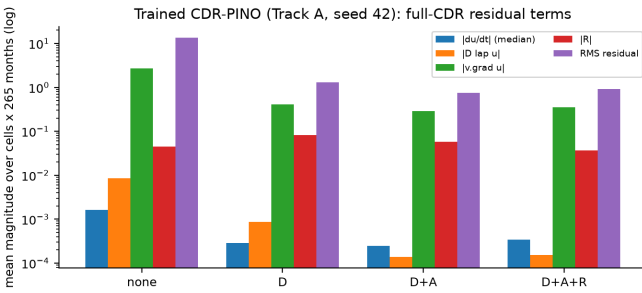


Fig. 8. Mean magnitudes of the rate of change, of each right-side term and of the residual for trained track A checkpoints (log scale).

night land surface temperature 14.3% (10.1%), air temperature 13.2% (13.1%), day land surface temperature 8.1% (9.6%) and slope 5.8% (5.6%). Specific humidity (3.1% against 13.0%) is not reproduced, and elevation is ranked higher here (10.7% against 2.4%) (Fig. 11 in the Appendix). The group totals reported in the reference study (33.9%, 26.1%, 10.8% and 9.7%) are permutation importances, although they are labelled as contributions. The pixel-level scores in Table III cannot be compared with the reference value, because the population, resolution, label and evaluation design all differ.

TABLE VII
PAIRED Δ ROC-AUC DECOMPOSITION (ALL / FOREST CELLS)

Source of difference	Δ ROC-AUC
RF minus MaxEnt, reference 15 predictors	+0.025 / +0.067
RF minus MaxEnt, 55 features	+0.008 / +0.031
55 features minus reference 15 (RF)	+0.005 / +0.012
55 features minus reference 15 (MaxEnt)	+0.022 / +0.048
55 features minus operator's 5 static covariates (RF)	+0.017 / +0.071
Trend group	+0.001 / +0.006
Land cover group	+0.003 / +0.008
Terrain group	-0.0001 / -0.0006
Physics terms, full minus none (operator)	-0.006 (A), n.s. (B1, B2), -0.011 (B3)
Operator minus RF, same cells and covariates	-0.041 (A)

I. National Susceptibility Map

Fig. 10 shows the track A random forest score at 1/120 degree, grouped into five classes by quantile breaks over forest cells (0.043, 0.210, 0.582 and 0.893). The map is distributed as a relative score and is tagged accordingly, since its mean (0.135) exceeds the prevalence (0.065) and no calibrated uncertainty estimate was established. High classes concentrate in the northeast, the central Indian highlands, the Eastern Ghats and the Himalayan foothills.

V. DISCUSSION

A. Why the Equation Did Not Help

Four observations explain the negative result. First, the target is static while the equation is dynamic. Susceptibility, as defined here and in the reference study, is a long-run property of a place, so the data loss rewards a stable ranking of cells and the simplest optimum is a nearly static field, which is what the trained models produce. For a static field the right side of (2) would have to vanish, and nothing in the data enforces this, so the residual is absorbed rather than reduced. Second, the terms describe spread, whereas monthly detections aggregated to 12 km record where fires occur, not how they move between cells

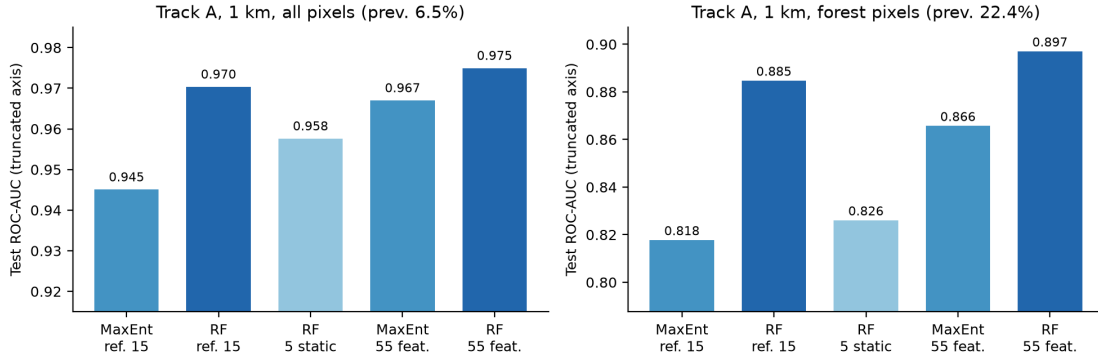


Fig. 9. Track A test ROC-AUC at 1 km for MaxEnt and random forest with the reference fifteen predictors (ref. 15), the operator's five static covariates (5 static) and the full 55 features (55 feat.), over all cells (left, prevalence 6.5%) and forest cells (right, prevalence 22.4%). The vertical axes are truncated.

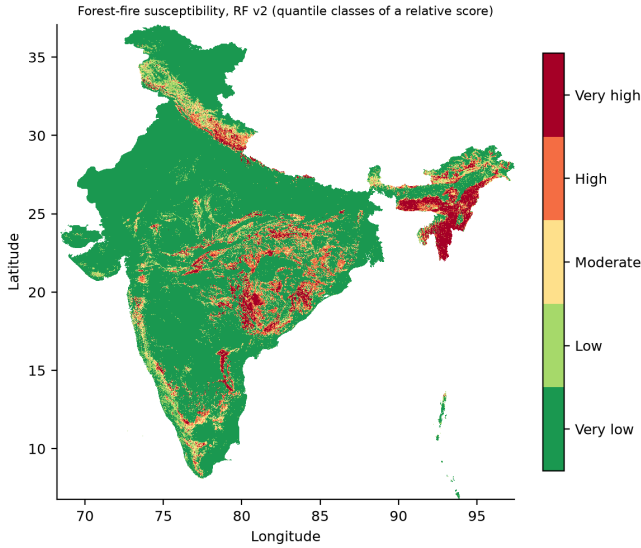


Fig. 10. Forest fire susceptibility of India from the random forest with 55 features, shown as five quantile classes of a relative score computed over forest cells.

within a month. The negligible share of diffusion is consistent with this. Third, advection behaves as a channel for elevation: it carries most of the right side in every configuration and recovers most of what diffusion alone loses, yet terrain adds nothing once the full predictor set is available. Fourth, the weak spatial transfer belongs to the operator and not to the inputs. With identical cells and covariates, even a linear model transfers far better across blocks and regions. A global Fourier representation trained on a single national domain may tie distant regions together and learn structure that does not carry over to unseen areas. The network without physics shares this weakness, so the physics terms neither cause nor cure it.

B. Implications for Physics-Informed Hazard Mapping

The design included the conditions under which physics-informed models are usually expected to help, namely spatial, regional and temporal shift, sparse positives and several seeds, and no benefit appeared. This does not rule out physics-informed learning for fire. Spread prediction, where the equation describes the observed process, is a different problem. For susceptibility applications, the results suggest four minimum reporting elements: the same network without the equation, classical models on identical cells and covariates, covariate-free persistence baselines for any temporal claim, and a consistency check of the trained field against its equation.

C. Lessons from the Pipeline

Several defects corrected here are generic. Rounding coordinates against a raster edge moved three quarters of the fire points by one cell. A time mean of anomalies over its own baseline is degenerate. The Mann-Kendall test applied to seasonal or smoothed series misstates significance counts by large factors: here the Seasonal Kendall test found 6.7 times (day) to 117 times (night) more significant cells for land surface temperature, and for air temperature it reversed the earlier conclusion. Reporting all-cell scores for a forest label inflates accuracy because non-forest cells are trivially negative. None of these corrections changed any model score by more than 0.012, so the conclusions above do not depend on them, but the descriptive statistics that relied on the earlier choices were not valid.

D. Limitations

Only one operator family and one equation were tested, so the results do not exclude local operators or equations fitted to spread data. The operator's loss weights, number of modes and window length were not tuned beyond weight decay and early stopping. A larger search could narrow, but is unlikely to reverse, a gap of 0.26 to 0.39 under spatial hold-out. The global receptive field lets the operator see the covariates of test cells, never their labels, and the dryness index is standardised over all months; both effects would favour the operator. The operator runs at 12 km and the classical models at 1 km, which

is why every operator comparison is made on the operator's own cells. The label records detected fire rather than burned area or severity, although annual forest-masked burned area correlates with the annual fire counts ($r = 0.93$). The random forest map is a relative score without calibrated uncertainty. Resolution independence and fine-tuning of the operator were not evaluated and are not claimed. SHAP values could not be computed in the software environment used, so permutation importance and partial dependence were used instead.

VI. CONCLUSION

We tested whether a convection-diffusion-reaction equation embedded in a neural operator improves forest fire susceptibility mapping over India, holding the architecture, inputs, grid and evaluation split fixed. The physics terms did not improve discrimination on any of four tracks and reduced it where the difference could be resolved. Classical learners on identical inputs outperformed the operator on every track, by 0.04 on random splits and by 0.26 to 0.39 under spatial and regional hold-out. On unseen years the operator fell below covariate-free persistence baselines, and the trained field was nearly static and did not satisfy its equation. The strongest model in this study is a random forest on a corrected set of 55 predictors, with a forest-cell ROC-AUC of 0.897 on a random split and 0.838 under spatial hold-out, and its advantage comes from the learner and the predictors. We recommend the controlled design and the consistency diagnostic as minimum practice for physics-informed hazard models, and release the corrected national pipeline, the reimplementing of the reference protocol (0.893 ± 0.009 against a reported 0.879), all per-cell predictions and all scripts.

DATA AVAILABILITY

The raw products are public (NASA FIRMS and LP DAAC, NASA GES DISC, ESA CCI and C3S, USGS SRTM and Geofabrik OpenStreetMap). The processing code for every step, the classical experiments, the operator training code (102 runs), the per-cell predictions, the analysis tables and the figures are available in the project repositories: https://github.com/rawan230/Spatial_Mapping_INDIA-2000-2022-, <https://github.com/rawan230/Integrated-Fire-Risk-Analysis-in-India-and-Impliment-Baseline-Random-Forest-Model-> and https://github.com/rawan230/_PINO_For_Spatial_Mapping_INDIA-2000-2022-, together with the linked step repositories.

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APPENDIX A SUPPLEMENTARY RESULTS

This appendix reports the permutation importance of the reimplemented reference protocol (Fig. 11), the sensitivity of MaxEnt to the number of training rows (Fig. 12) and the remaining sensitivity analyses (Table VIII). All values are read from the result files released with this study.

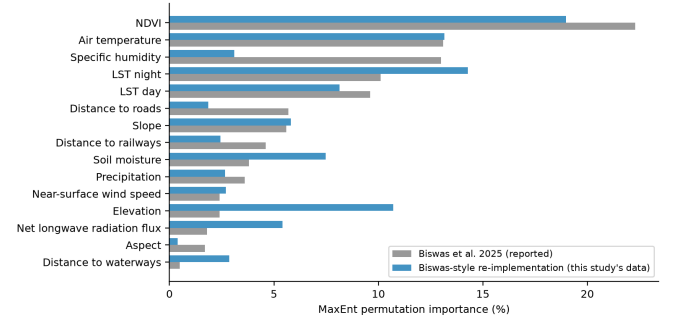


Fig. 11. Permutation importance of the reimplemented reference protocol compared with the values reported in [1].

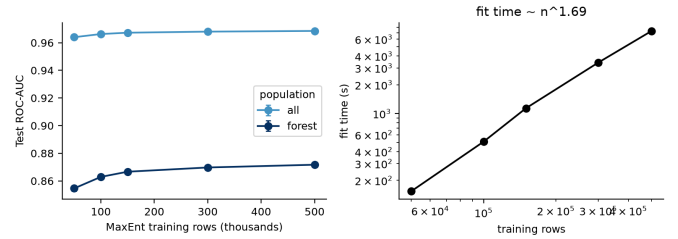


Fig. 12. MaxEnt test ROC-AUC and fitting time as a function of the number of training rows (50 000 to 500 000).

TABLE VIII
SENSITIVITY ANALYSES (RANDOM FOREST, TRACK A UNLESS STATED)

Factor	Effect on ROC-AUC
FIRMS confidence ≥ 30 and type 0 filters	< 0.001
NDVI quality level (good only vs good and marginal)	negligible
LST aggregated to 0.05° (MOD11C3 resolution)	≤ 0.0002
Slope algorithm (Horn vs Zevenbergen-Thorne)	0.0000
Slope from a 0.25° DEM	negligible; mean slope 0.29° against 5.72°
Held-out year label built with test years (operator)	+0.0006
MaxEnt rows 50k to 500k (all / forest)	0.964 to 0.969 / 0.855 to 0.872; fit time grows as about $n^{1.6}$

REFERENCES

- [1] U. Biswas, S. Mahato, and P. K. Joshi, "Spatial prediction of forest fires in India: a machine learning approach for improved risk assessment and early warning systems," *Environmental Science and Pollution Research*, vol. 32, no. 8, pp. 4856–4878, 2025.
- [2] A. R. Uthappa, B. Das, A. Raizada, P. Kumar, P. Jha, and P. V. V. Prasad, "Forest fire susceptibility mapping using multi-criteria decision making and machine learning models in the Western Ghats of India," *Journal of Environmental Management*, vol. 379, p. 124777, 2025.
- [3] Kanda Naveen Babu, R. Gour, Kurian Ayushi, N. Ayyappan, and N. Parthasarathy, "Environmental drivers and spatial prediction of forest fires in the Western Ghats biodiversity hotspot, India: An ensemble machine learning approach," *Forest Ecology and Management*, vol. 540, p. 121057, 2023.
- [4] G. Meraj, S. Hashimoto, R. Dasgupta, and B. K. Mitra, "Ecological risk assessment and management of forest fires in Tamil Nadu, India: A MaxEnt model-based approach for strategic resource allocation and fire mitigation," *Risk Analysis*, vol. 45, no. 11, pp. 3604–3625, 2025.

- [5] R. Guria, M. Mishra, S. Mohanta, and S. Paul, "Forest fire probability zonation using dNBR and machine learning models: a case study at the Similipal Biosphere Reserve (SBR), Odisha, India," *Environmental Science and Pollution Research*, vol. 32, no. 59, pp. 31 375–31 396, 2025.
- [6] P. Gupta, A. K. Shukla, and D. P. Shukla, "Machine learning-based forest fire susceptibility mapping of Southern Mizoram, a part of Indo-Burma biodiversity hotspot," *Environmental Science and Pollution Research*, vol. 32, no. 59, pp. 31 433–31 454, 2025.
- [7] M. S. Sarkar, B. K. Majhi, B. Pathak, T. Biswas, S. Mahapatra, D. Kumar, I. D. Bhatt, J. C. Kuniyal, and S. Nautiyal, "Ensembling machine learning models to identify forest fire-susceptible zones in Northeast India," *Ecological Informatics*, vol. 81, p. 102598, 2024.
- [8] H. T. Hang, S. Mallick, S. Alqadhi, A. A. Bindajam, and H. G. Abdo, "Exploring forest fire susceptibility and management strategies in Western Himalaya: Integrating ensemble machine learning and explainable AI," *Environmental Technology & Innovation*, vol. 35, p. 103655, 2024.
- [9] F. A. Malik, F. Mushtaq, M. Farooq, L. T. S. Guite, S. Kanga, G. Meraj, S. K. Singh, and P. Kumar, "Assessing forest fire vulnerability with fuzzy-AHP: insights from Poonch forest division, Jammu and Kashmir," *Discover Forests*, vol. 1, no. 1, p. 4, 2025.
- [10] G. Zhang, M. Wang, and K. Liu, "Forest fire susceptibility modeling using a convolutional neural network for Yunnan province of China," *International Journal of Disaster Risk Science*, vol. 10, no. 3, pp. 386–403, 2019.
- [11] O. Kantarcioğlu, K. Schindler, and S. Kocaman, "Forest fire susceptibility assessment with machine learning methods in North-East Türkiye," in *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLVIII-M-1-2023, 2023, pp. 161–167.
- [12] M. C. İban and O. Aksu, "SHAP-driven explainable artificial intelligence framework for wildfire susceptibility mapping using MODIS active fire pixels: A case study in Izmir, Türkiye," *Remote Sensing*, vol. 16, no. 15, p. 2842, 2024.
- [13] R. Y. Zakari, O. A. Malik, and W.-H. Ong, "Machine learning-driven wildfire susceptibility mapping in New South Wales, Australia using remote sensing and explainable artificial intelligence," *Natural Hazards*, vol. 121, no. 13, pp. 15 331–15 357, 2025.
- [14] P. Symeonidis, T. Vafeiadis, D. Ioannidis, and D. Tzovaras, "Wildfire susceptibility mapping in Greece using ensemble machine learning," *Earth*, vol. 6, no. 3, p. 75, 2025.
- [15] K. Gholamnia, H. Tahmasebi Moghaddam, G. Einali, B. Akbari Monfared, G. Lorestani, O. Ghorbanzadeh, and J. Einali, "Uncertainty-aware machine learning via Dempster-Shafer theory for wildfire susceptibility mapping," *Spatial Information Research*, vol. 34, no. 4, p. 35, 2026.
- [16] V. P. Santana Neto, A. J. N. Nunes, F. T. P. Torres, J. M. Gleriani, and D. N. Cosenza, "Assessing wildfire susceptibility and driving variables in Portugal using machine learning approach," *Journal for Nature Conservation*, vol. 86, p. 126956, 2025.
- [17] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillerá-Arroita *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.
- [18] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
- [19] Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Aziz-zadenesheli, and A. Anandkumar, "Physics-informed neural operator for learning partial differential equations," 2023, arXiv:2111.03794.
- [20] P. Jiang, Z. Yang, J. Wang, C. Huang, P. Xue, T. C. Chakraborty, X. Chen, and Y. Qian, "Efficient super-resolution of near-surface climate modeling using the Fourier neural operator," *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 7, p. e2023MS003800, 2023.
- [21] A. Y. Sun, P. Jiang, P. Shuai, and X. Chen, "Bridging hydrological ensemble simulation and learning using deep neural operators," *Water Resources Research*, vol. 60, no. 10, p. e2024WR037555, 2024.
- [22] T. Kurth, S. Subramanian, P. Harrington, J. Pathak, M. Mardani, D. Hall, A. Miele, K. Kashinath, and A. Anandkumar, "FourCastNet: Accelerating global high-resolution weather forecasting using adaptive Fourier neural operators," in *Proc. Platform for Advanced Scientific Computing Conf. (PASC '23)*, 2023.
- [23] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021.
- [24] J. S. Read, X. Jia, J. Willard, A. P. Appling, J. A. Zwart, S. K. Oliver, A. Karpatne, G. J. A. Hansen, P. C. Hanson, W. Watkins, M. Steinbach, and V. Kumar, "Process-guided deep learning predictions of lake water temperature," *Water Resources Research*, vol. 55, no. 11, pp. 9173–9190, 2019.
- [25] K. Vogiatzoglou, C. Papadimitriou, V. Bontozoglou, and K. Ampountolas, "Physics-informed neural networks for parameter learning of wildfire spreading," *Computer Methods in Applied Mechanics and Engineering*, vol. 434, p. 117545, 2025.
- [26] J. J. Dabrowski, D. E. Pagendam, J. Hilton, C. Sanderson, D. MacKinlay, C. Huston, A. Bolt, and P. Kuhnert, "Bayesian physics informed neural networks for data assimilation and spatio-temporal modelling of wildfires," *Spatial Statistics*, vol. 55, p. 100746, 2023.
- [27] L. Giglio, W. Schroeder, and C. O. Justice, "The collection 6 MODIS active fire detection algorithm and fire products," *Remote Sensing of Environment*, vol. 178, pp. 31–41, 2016.
- [28] B. K. P. Horn, "Hill shading and the reflectance map," *Proceedings of the IEEE*, vol. 69, no. 1, pp. 14–47, 1981.
- [29] R. M. Hirsch, J. R. Slack, and R. A. Smith, "Techniques of trend analysis for monthly water quality data," *Water Resources Research*, vol. 18, no. 1, pp. 107–121, 1982.
- [30] P. K. Sen, "Estimates of the regression coefficient based on Kendall's tau," *Journal of the American Statistical Association*, vol. 63, no. 324, pp. 1379–1389, 1968.
- [31] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.
- [32] L. Anselin, "Local indicators of spatial association: LISA," *Geographical Analysis*, vol. 27, no. 2, pp. 93–115, 1995.
- [33] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [34] S. J. Phillips, R. P. Anderson, and R. E. Schapire, "Maximum entropy modeling of species geographic distributions," *Ecological Modelling*, vol. 190, no. 3–4, pp. 231–259, 2006.
- [35] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.
- [36] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.